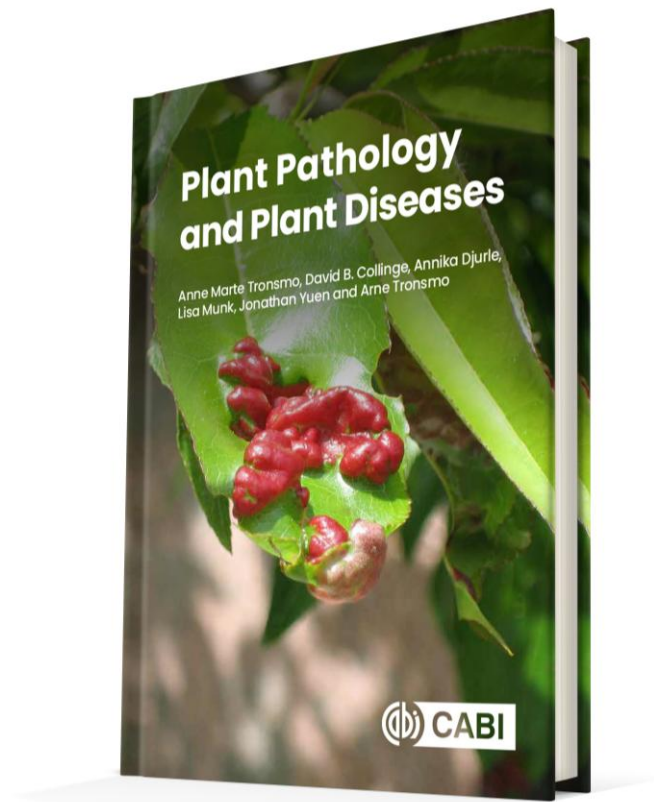




Plant Pathology and Plant Diseases

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Lisa Munk, Jonathan Yuen and Arne Tronsmo 2020

Fig. 9.1 A-B Examples of symptoms. (A) *Rhytisma acerinum* causing tar spot on acer. (B) Canker of apple tree caused by *Neonectria ditissima*. (A, B, © L. Munk.)



Fig. 9.1 C-D Examples of symptoms. (C) *Colletotrichum gloeosporioides* causing bitter rot on apple. (D) *Aphanomyces cochlioides* causing dry rot of sugar beet. (C: © L. Munk; D: © A. Djurle.)



Fig. 9.1 E-F Examples of visible signs of disease. (E) Smut spores (teliospores) of *Ustilago maydis*. (F) Mycelium and sclerotia of *Sclerotinia sclerotiorum*. (© L. Munk.)



Fig. 9.2 A-B Examples of visible signs of disease. (A) Conidia of *Monilia fructigena* causing brown rot on apple. (B) Conidia and chasmothecia of *Erysiphe trifolii* causing powdery mildew on lupin.. (© L. Munk).



Fig. 9.2 C-D Examples of visible signs of disease. (C) Aecia of *Uromyces beticola* causing beet rust. (D) Uredinia of *Puccinia striiformis* causing yellow rust on wheat. (© L. Munk).



Fig. 9.2 E-F Examples of visible signs of disease. (E) Smut spores (teliospores) of *Ustilago maydis*. (F) Mycelium and sclerotia of *Sclerotinia sclerotiorum*. (© L. Munk.)



Fig. 9.2 G-H Examples of visible signs of disease. (G) Sporangia of *Albugo candida* causing white rust on shepherd's purse. (H) Sporangia of *Peronospora viciae* causing downy mildew on faba bean. (© L. Munk.)

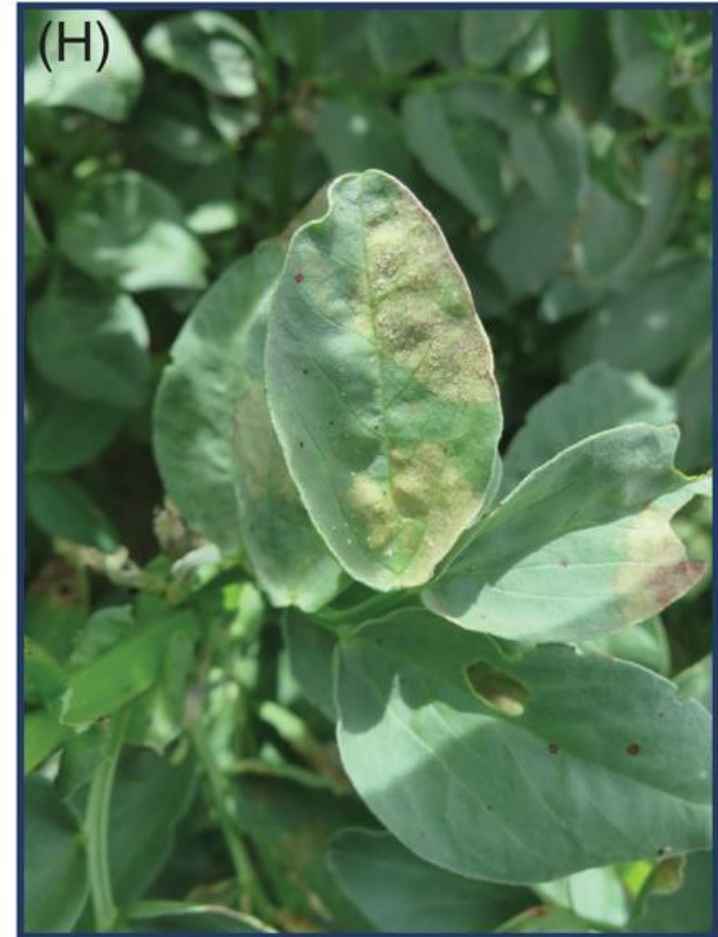


Fig. 9.3 A-B Examples of abiotic disorders. (A) Hail damage on oilseed rape. (B) Frost damage on a shoot of grapevine.
(A: © SLU; B: © L. Munk.)



Fig. 9.3 C-D Examples of abiotic disorders. (C) Ice damage on golf course. (D) Kale with leaf drop due to sudden change in temperature. (C: © A. Tronsmo. D: © L. Munk.)



Fig. 9.3 E-F Examples of abiotic disorders. (E) Iron deficiency. (F) Boron deficiency in celeriac. (E: T. Krogstad, © NMBU F: I. Aasen, © NMBU.)



Fig. 9.4 A compound microscope with phase-contrast. 1, Eye pieces; 2, rotating revolver with 4 objectives; 3, stage; 4, condenser; 5, ring for shift between light field, phase-contrast and dark field; 6, adjustment knob for the aperture; 7, adjustment knob for condenser; 8, coarse and fine adjustment knob; 9, lamp; 10, power switch; 11, light control; 12, brightness adjustment. (© A. Tronsmo)

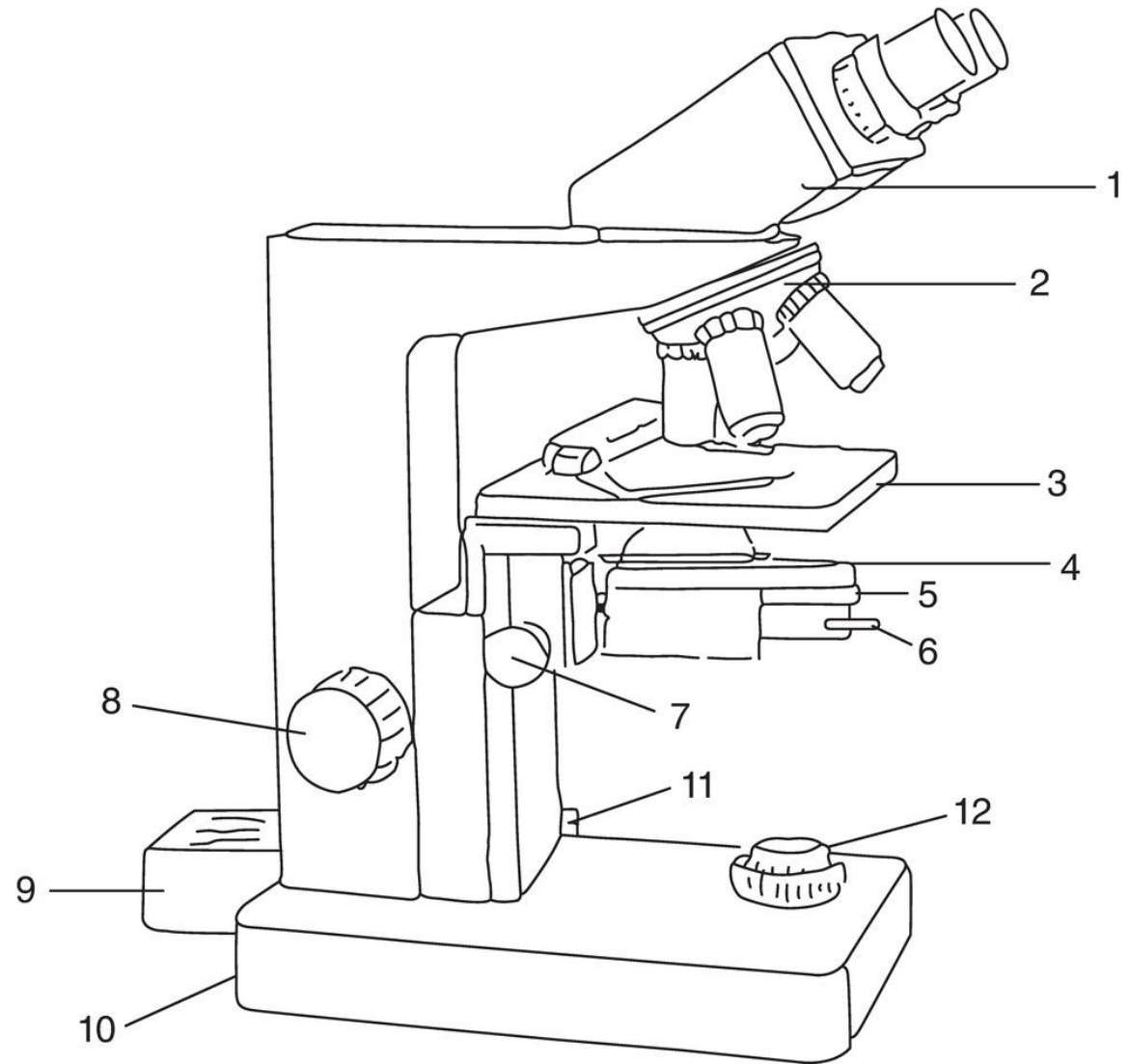


Fig. 9.5 A-B Examples of fungal and fungal-like structures. (A) Spores of *Fusarium* sp. (B) A conidiophore and conidia of *Botrytis cinerea*. (E) Mature and immature teliospores of *Phragmidium* sp. (F) Sporangiophores and a sporangium of *Peronospora viciae* on pea. (© L. Munk).

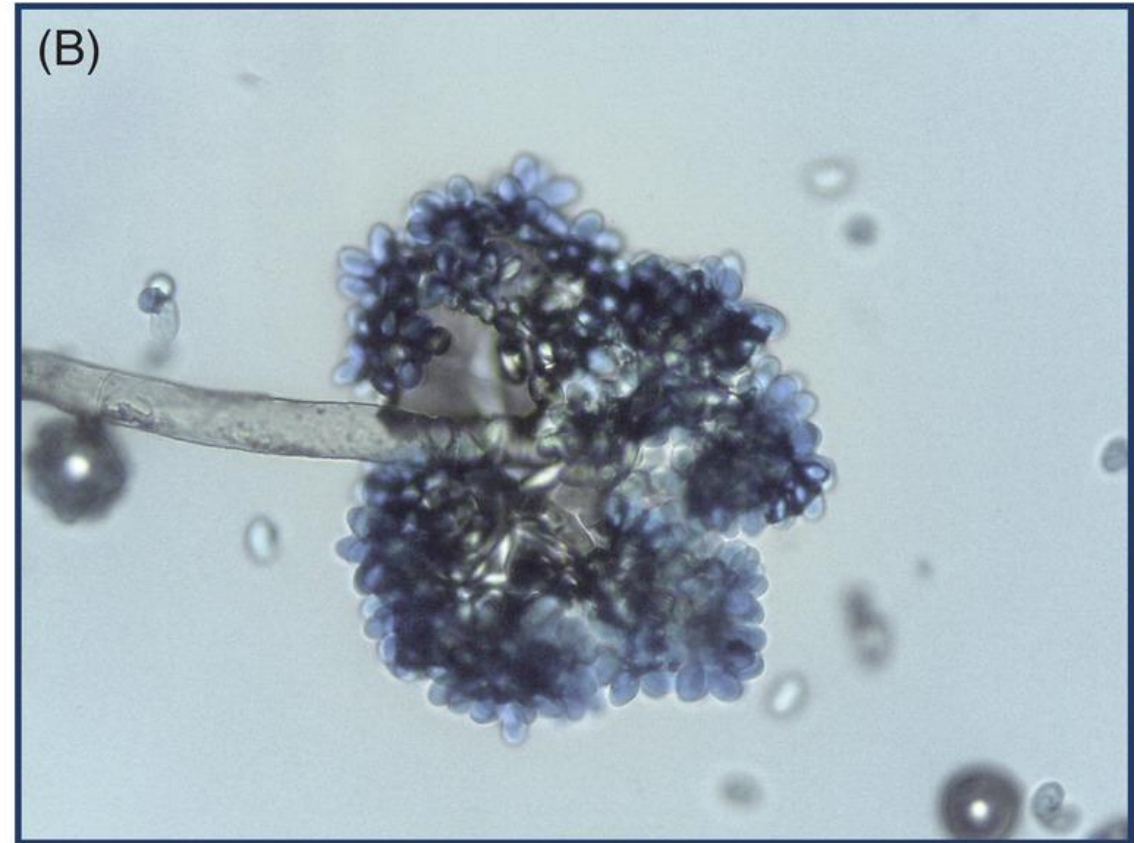


Fig. 9.5 C-D Examples of fungal and fungal-like structures. (C) Ascospores of *Venturia inaequalis*. (D) Chasmothecium – ascocarp of the fungus causing powdery mildew on oak, *Erysiphe alphitoides*. (© L. Munk.)

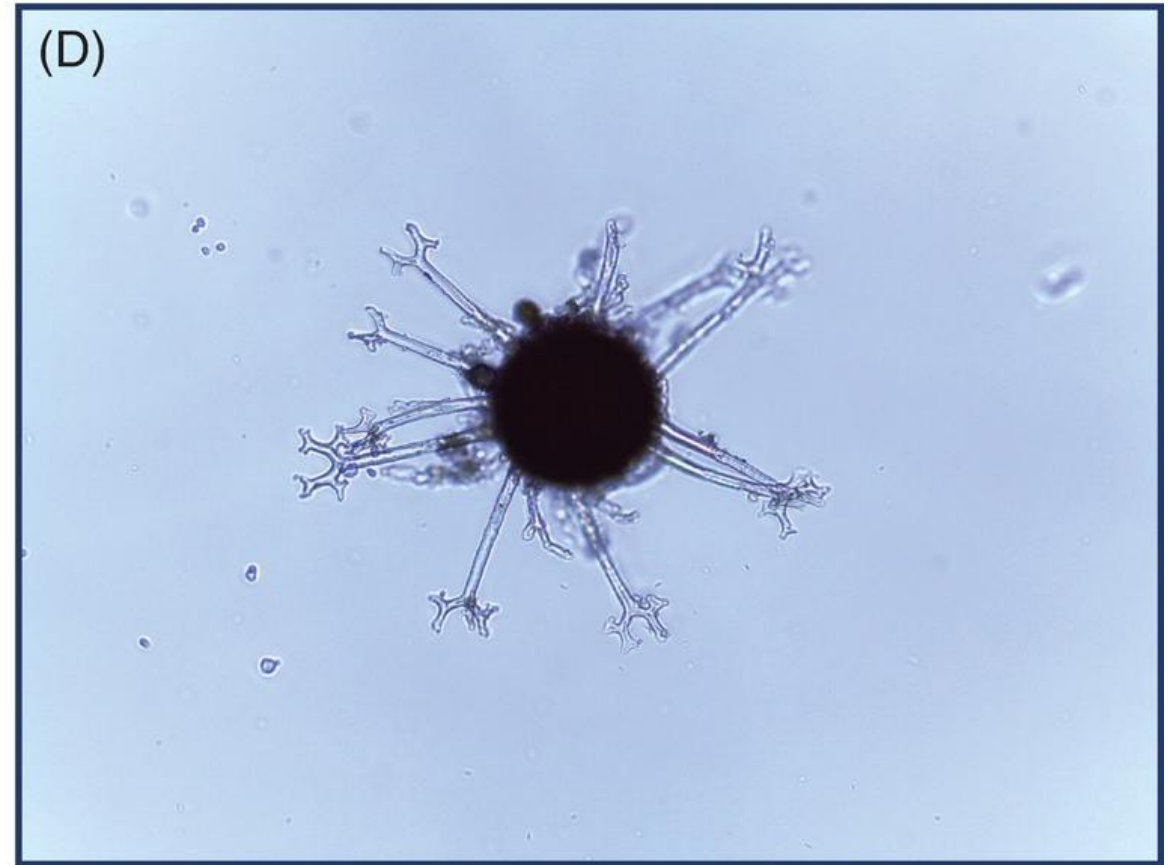


Fig. 9.5 E-F Examples of fungal and fungal-like structures. (E) Mature and immature teliospores of *Phragmidium* sp. (F) Sporangioophores and a sporangium of *Peronospora viciae* on pea. . (© L. Munk).

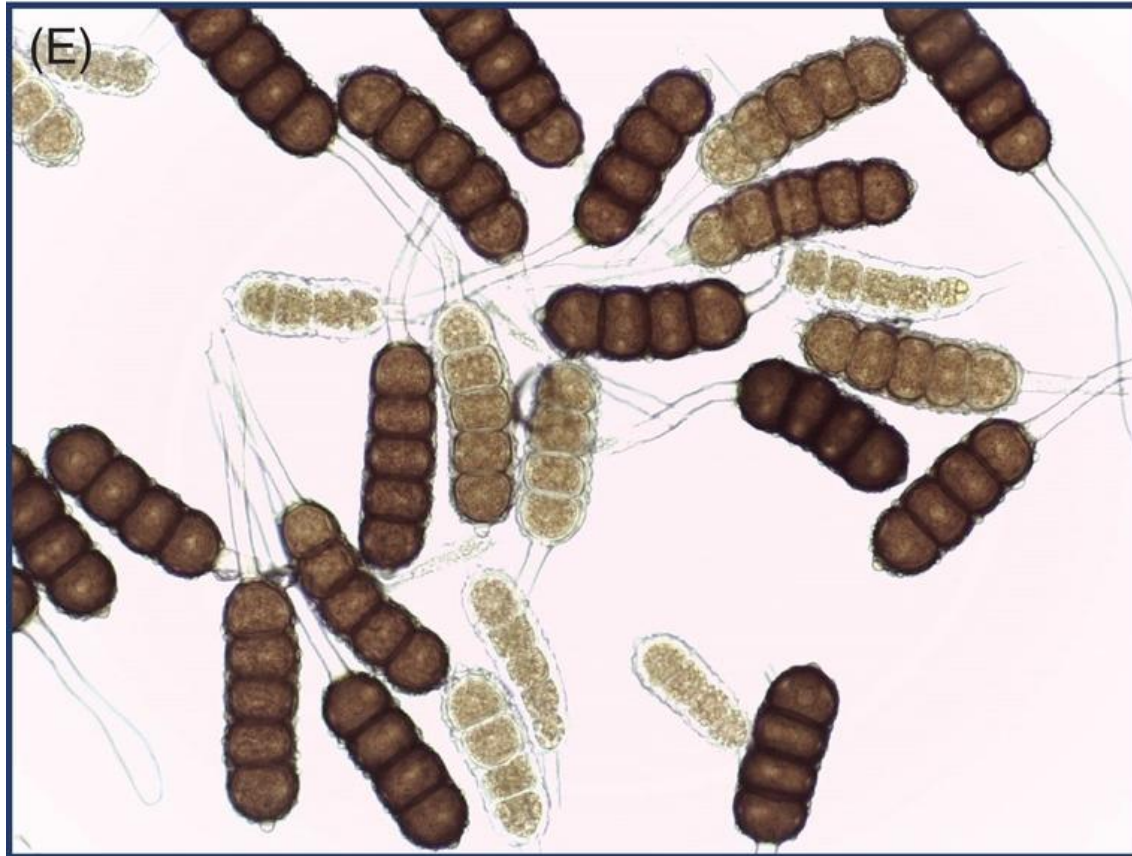


Fig. 9.6 A scheme for diagnosis following isolation of the pathogen. (© L. Munk.)

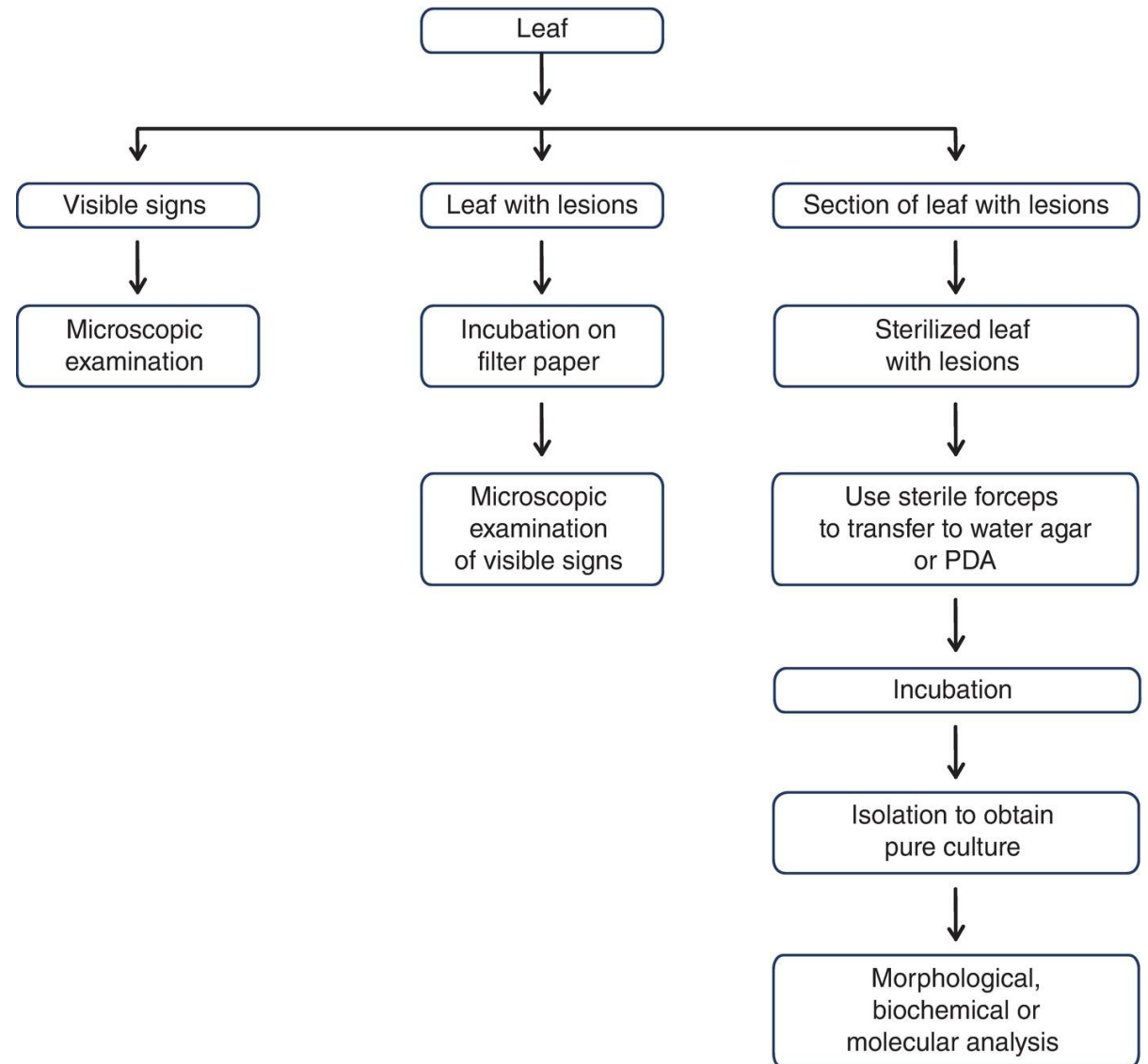
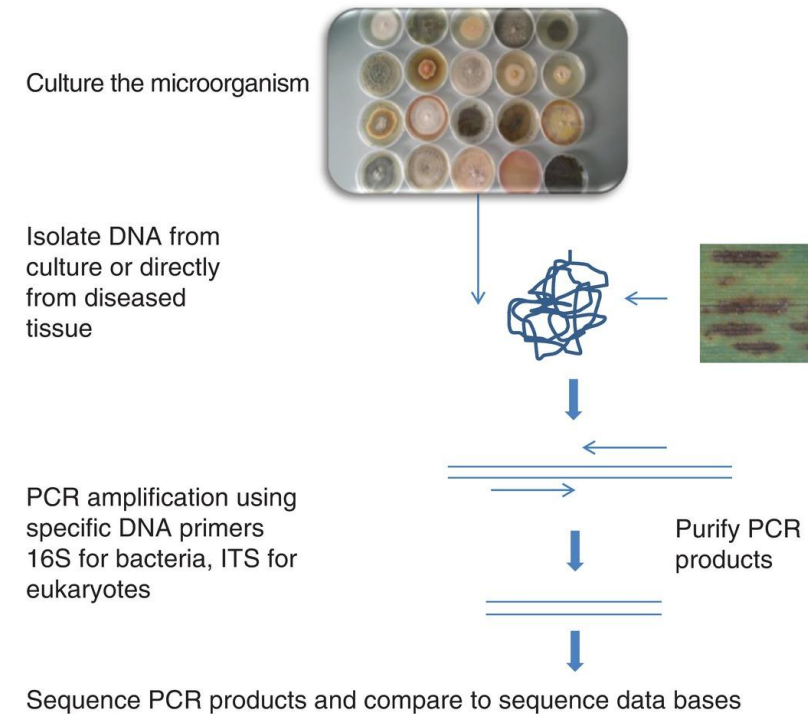


Fig. 9.7 DNA sequencing for identification of microorganisms: (A) Scheme for isolating DNA and PCR amplification of specific DNA sequences from infected plant tissue. (B) Schematic structure of eukaryotic ribosomal genes illustrating the internal transcribed spacer (ITS) sequences. (© D.B. Collinge).

(A) Scheme for sequencing protocol



(B) Schematic structure of eukaryotic ribosomal genes illustrating the internal transcribed spacer sequences ITS

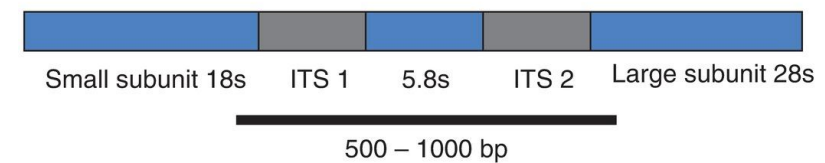


Fig. 9.8 Lateral flow device – field kit. (© L. Munk.)

